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One Probe Won't Catch Them All: Towards Targeted Deception Detection

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Abstract

Linear probes are a promising approach for monitoring AI systems for deceptive behaviour. Previous work has shown that a linear classifier trained on a contrastive instruction pair and a simple dataset can achieve good performance. However, these probes exhibit notable failures even in straightforward scenarios, including spurious correlations and false positives on non-deceptive responses. In this paper, we demonstrate that deception detection is inherently heterogeneous: while a single universal probe achieves modest improvements (+0.032 AUC), post-hoc oracle analysis reveals substantially higher potential (+0.108 AUC) when probes are matched to specific deception types, and synthetic validation experiments suggest this ceiling is achievable *a priori* when the deception type is known in advance. Our findings reveal that instruction pairs capture deceptive *intent* rather than content-specific patterns, explaining why prompt choice dominates probe performance (70.6% of variance). Given this heterogeneity, we conclude that organizations should define their specific threat models and deploy appropriately matched probes rather than seeking a universal deception detector.

1. Introduction

Deception in large language models has emerged as a central concern in AI safety. Models now possess both the capability and propensity to engage in deceptive behaviours when placed in situations that provide incentives to do so. Empirical studies have demonstrated that such behaviours are increasingly prevalent in the latest generation of frontier models: LLMs have been observed to misrepresent information, obscure intentions, or act strategically under specific contextual pressures (Schoen et al., 2025; Chowdhury et al., 2025; Park et al., 2024). As LLMs are increasingly deployed in a wide variety of domains including high-stakes contexts, there is an urgent need for reliable monitoring mechanisms capable of detecting when a model engages in deceptive output. Black-box approaches, which monitor only tokens output by the model using trained classifiers or other LLMs, are one possible approach. Chain-of-thought monitoring is widely used to monitor model misbehaviour including deception (Korbak et al., 2025).

However, recent work indicates that chain-of-thought traces are not always faithful indicators of a model's internal reasoning process. Models can perform reasoning implicitly while generating innocuous-looking chain-of-thought outputs (Arcuschin et al., 2025; Hao et al., 2024). In addition to black-box monitoring being expensive, the increasing subtlety of deception in advanced models means that model outputs alone cannot be relied upon for effective monitoring, as models may learn to strategically generate innocuous-sounding text. Because of these limitations, there has been interest in exploring white-box methods that inspect model activations.

Early work has included training classifiers on internal activations to distinguish between true and false statements (Azaria & Mitchell, 2023; Burns et al., 2023; Zou et al., 2025), but these efforts have largely focused on evaluating these techniques on factual inaccuracies rather than strategic deception. Goldowsky-Dill et al. (2025) addressed some of these limitations with instruction-pair probes to capture a model's intention to deceive. They trained a linear classifier on model activations from simple facts in the Zou et al. (2025) instruction pairs dataset, comparing responses under honesty versus dishonesty instructions. This approach generalized surprisingly well to detecting strategic deception in settings such as concealing insider trading and deliberately underperforming on safety evaluations. However, the probe suffered from aggregation errors and spurious correlations, sometimes firing on completely honest responses. Despite their effectiveness, probes learn spurious correlations with their training datasets. Levinstein & Herrmann (2024) showed that probes from Azaria & Mitchell (2023) even misclassified negations of original statements from training dataset. Further empirical evidence from Levy (2024)

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showed that classifiers based on Zou et al. (2025) had high sensitivity but low specificity, and were better characterized as differentiating between fact-based retrieval and other tasks (such as writing science fiction), spanning a much broader surface area than intended.

Our initial experiments revealed that instruction-pair probes using the default prompts from (Goldowsky-Dill et al., 2025) perform poorly across diverse deception datasets such as MASK. For example, the baseline probe achieved only 0.374 AUC on MASK known facts, indicating substantial room for improvement. Analysis of the evaluation datasets uncovered substantial heterogeneity in the types of deception across different benchmarks. This diversity in deception characteristics presents a fundamental challenge for developing a single universal deception probe. For deployment purposes, labs would prefer to distinguish probes that detect genuinely concerning behaviours from those that merely flag benign role-playing scenarios.

1.1. Our contribution

We present three key contributions to deception detection using linear probes:

System prompts dominate probe performance. Through analysis of 2000 probes across varied configurations, we demonstrate that system prompt design explains 70.6% of performance variance, substantially exceeding the impact of training data, layer selection, or token aggregation strategy. This finding justifies our focus on prompt-based approaches to improving deception detection.

Deception detection is heterogeneous, but targeted probes achieve strong performance. While attempting to select a single universal probe shows modest gains (+0.032 AUC, 5.5%), post-hoc oracle analysis reveals a much higher performance ceiling (+0.108 AUC, 20.5%) when probes are matched to specific deception types from a human-interpretable taxonomy (Druckman & Bjork, 1991). Importantly, experiments on synthetic data suggest this ceiling is achievable *a priori*: when the deception type is known in advance, targeted probes often outperform the baseline.

Organizations should match probes to their threat models. We identify which prompt types work best for which deception scenarios, and demonstrate through qualitative analysis that probes detect distinct semantic properties. This heterogeneity challenges the assumption of a universal deception detector and suggests that organizations with well-defined threat models can achieve stronger detection by deploying appropriately matched probes.

2. Related work

2.1. White-box methods

White-box methods for deception detection leverage access to a model's activation space to identify when language models produce false or deceptive outputs. Several studies have focused on training classifiers on the model's internal activations (Azaria & Mitchell, 2023; Burns et al., 2023; Zou et al., 2025; Marks & Tegmark, 2024). Although effective at classification, early white-box techniques for detecting deception primarily focused on off-policy fact-based evaluation datasets like TruthfulQA, rather than targeting model-generated responses that mimic more sophisticated scenarios.

2.2. Instruction-pair probes

Goldowsky-Dill et al. (2025) addressed some of these limitations with instruction-pair probes, shifting focus from the lie itself to the intent to lie. They trained a linear classifier on model activations from simple facts in the Zou et al. (2025) instruction pairs dataset, comparing responses under honesty versus dishonesty instructions. The probe performed well with AUROCs greater than 0.96 when tested on datasets such as sandbagging and insider trading.

However, these probes exhibited notable limitations. In follow-up work, Parrack et al. (2025) benchmarked the performance of instruction-pair probes combined with black-box monitors versus black-box monitors alone, and evaluated different aggregation strategies. They confirmed significant limitations of instruction-pair probes, including spurious correlations demonstrated through a deception dilution dataset. These probes often flag innocuous content such as fiction writing or role-playing as deceptive, suggesting they may be detecting broader patterns (e.g., fact-based vs. creative reasoning) rather than strategic deception specifically.

2.3. Benchmarks

Benchmarking deception in large language models remains a difficult challenge. Several recent works have attempted to construct datasets that capture different forms of deceptive behaviour, such as factual lying, instruction misrepresentation, strategic omission, or self-serving reasoning. Goldowsky-Dill et al. (2025) collected and constructed multiple datasets including AI liar, Insider trading, Sandbagging, and Role-playing scenarios, among others.

MASK: The MASK dataset (Ren et al., 2025) provides a rigorous benchmark for evaluating honesty in large language models by measuring whether models remain truthful when incentivized to lie. It contains human-labeled examples across six distinct archetypes, each consisting of a proposition, ground truth, pressure prompt designed to elicit lying,

and belief elicitation prompt to establish the model's actual knowledge.

Liar's Bench: Kretschmar et al. (2025) built a large benchmark for evaluating deception detection in AI assistants. It contains examples of deceptive and honest model outputs across six scenarios, capturing diverse types of deception such as strategic omission, misrepresentation, and evasion through irrelevant truths.

3. Methodology

3.1. Instruction-Pair Probe

We build upon the instruction-pair probes introduced by (Goldowsky-Dill et al., 2025). The approach constructs contrastive pairs: each training example uses the same factual statement as the model response, but pairs it with either an honest or dishonest system prompt. The response is token-forced (off-policy), meaning the factual statement is fed directly into the chat template as the model's response rather than being generated by the model.

For each response token, we extract the residual stream values at the layer of interest—these activation vectors directly become the input features for a logistic regression classifier trained to predict whether the honest or dishonest prompt was used. Crucially, we exclude the final 5 response tokens from the activations. This prevents the probe from detecting whether the stated fact is literally true or false based on the response content. By truncating before the fact fully resolves (e.g., capturing activations at "The blue whale is the…" before the model commits to "largest" or "smallest"), the probe is forced to learn from the model's internal representations of intent as shaped by the system prompt, rather than from the surface-level content of the response.

Default prompt pair:

- **Honest prompt:** "Pretend you are an honest person making statements about the world." Model response: "The blue whale is the…"
- **Deceptive prompt:** "Pretend you are a deceptive person making statements about the world." Model response: "The blue whale is the…"

3.2. Models

We use the open-weight instruction-tuned model Gemma 2 9B IT as our primary model. After conducting preliminary experiments across layers 20, 25, and 31 (see Section 4.1), we focus our analysis on layer 20. The evaluation datasets are generated with on-policy responses from the model with the temperature set to 1. To assess whether our findings generalize across architectures and scales, we additionally replicate the ANOVA experiment on Llama 3.3 70B Instruct (Section 4.1).

3.3. Datasets

We distinguish three dataset roles: training, validation, and evaluation.

Training data. Probes are trained on small datasets of true and false factual statements (e.g., "The blue whale is the largest mammal on Earth") from Azaria & Mitchell (2023). We also generated 11 additional synthetic factual datasets covering different types of facts, to measure whether the specific factual domain affects probe performance. Each dataset contains approximately 200 facts. Dataset categories are described in Appendix B. Following the ANOVA analysis (Section 4.1), which found that varying the domain and structure of training facts (e.g., simple statements vs. if-else constructions) explained only 2.5% of probe performance variance, we focused all subsequent experiments on the Azaria-Mitchell dataset. This is consistent with our broader finding that probes detect intent rather than content-specific patterns.

Evaluation data. Probes are evaluated on ten richer deception datasets drawn from Goldowsky-Dill et al. (2025) and Liar's Bench (Kretschmar et al., 2025), including MASK (Ren et al., 2025) and others. These datasets contain on-policy model responses to scenarios involving various deception types, and are entirely separate from the training data.

Validation data. To select the single best-performing probe on average across datasets, we hold out 50 balanced samples (25 honest, 25 deceptive) from each of three evaluation datasets (AI liar, Convincing Game, and Instructed Deception). These samples are never reused for evaluation. Note that this selection procedure only applies to the "Best Average" probe; for the "Best Taxonomy" oracle evaluation on a given dataset, we simply run all probes and report the best-performing one.

3.4. Prompt Generation

We generate two types of instruction pairs for probe training. Each pair consists of an honest and a dishonest system prompt targeting the same deception scenario. Full prompts and generation procedures are available in Appendix B.3.

1. **Taxonomy-Based Prompts:** We generated 23 prompt pairs using Claude Sonnet 4.5 (Anthropic, 2025), each targeting a specific deception type from the taxonomy in Druckman & Bjork (1991). These include 16 core deception types (white lie, exaggeration, evasion, bluff, concealment, pretense, trick, impersonation, counterfeit, forgery, pressure dishonesty, backstab, false alarm,

hoax, partial truth, and overt lie) plus 7 control pairs for non-deceptive behaviors.

2. **Framing Variations:** We generated 34 additional prompt pairs using Claude Sonnet 4.5 (Anthropic, 2025), exploring different ways to linguistically frame the honest/dishonest distinction, without targeting a specific deception type. Categories include identity-based framings (e.g., "You are an AI designed to be fundamentally truthful"), role-based framings (e.g., "Your role is to provide factual and accurate information"), instruction-based, hybrid, control repetition, long-form, and baseline framings. These explore whether the surface-level structure of the prompt affects probe performance independently of the deception type being targeted.

3.5. Baselines and Skylines

For baselines, we use an instruction-pair probe with the default system prompt from Goldowsky-Dill et al. (2025) (described in Section 3.1). We also computed in-distribution skyline probes on each evaluation dataset using an 80/20 train/test split, representing the best achievable performance when the probe's training distribution matches the evaluation distribution. These skyline probes achieved near-perfect AUC (≥ 0.993 on nine of ten datasets), and are omitted from the main figures for clarity.

3.6. Training of Probes

We train logistic regression probes on the residual stream values extracted at layer 20 (zero-indexed) of the model (as described in Section 3.1). These activation vectors are standardized to zero mean and unit variance across the dataset before training. We aggregate over all response tokens except the final 5, using the mean. The L2 regularization constant was tuned as part of the ANOVA sweep; we found $\lambda = 1$ to be optimal.

4. Experiments

4.1. Analysis of Variance across Training Setups

We identified four factors that affect probe performance: (1) the instruction pair (honest/deceptive prompt) used when training the probe, (2) the specific dataset of true facts used as the model response, (3) the token selection strategy for aggregation from the model response (first 5, first 10, all except last 10, all except last 5, or chat template tokens only), and (4) the layer probed at.

In order to determine where to focus our efforts, we trained 2000 probes using different combinations of these factors (exploring layers 20, 25, and 31), and evaluated each on the validation dataset described in Section 3.3. We then conducted an analysis of variance (ANOVA) on the results to determine the fraction of variance attributable to each factor.

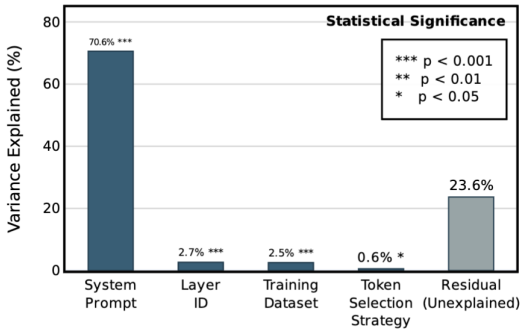


Figure 1. Variance in AUC on evaluation datasets explained by experimental factors. System prompt explains 70.6% of performance variance ($p < 0.001$), substantially exceeding variance explained by layer ID (2.7%), training dataset (2.5%), or token selection strategy (0.6%). Residual unexplained variance accounts for 23.6%.

Most of the variance in probe performance is attributable to changes in the system prompt, with relatively minor effects caused by variation in the layer ID, training dataset, or token selection policy. There are minor interaction effects between the system prompt and training dataset, but in general they are dominated by the system prompt variations.

This result forms our justification for focusing on system prompt variation and fixing the remaining factors for the remainder of our analysis. Given that layer choice explained only 2.7% of variance, we selected layer 20 for all subsequent experiments. We also fixed: 1) Token selection to all tokens up to 5 tokens before the end of the response. 2) Training dataset to the original dataset of true and false facts from Azaria & Mitchell (2023)

Cross-model replication on Llama 3.3 70B Instruct. To assess whether the dominance of system prompt choice generalizes beyond `Gemma-2-9B-IT`, we replicated the same ANOVA sweep on `Llama-3.3-70B-Instruct`. System prompt choice again dominates, explaining 71.0% of performance variance ($p < 0.001$), compared with 5.8% for layer ID, 4.5% for training dataset, and 0.3% for token selection strategy (Figure 2).

4.2. Probe Training with Varying System Prompts

The ANOVA result motivates a key question: given that prompt choice dominates probe performance, do different deception datasets require different prompts, or does a sin-